

```
class ImagePreprocessor:
    def __init__(self, img_size=512):
        self.img_size = img_size
        self.mean = np.array([0.485, 0.456, 0.406])
        self.std = np.array([0.229, 0.224, 0.225])

    def process(self, img_path):
        self.origin_img = cv2.imread(img_path)
        self.origin_img = cv2.cvtColor(self.origin_img, cv2.COLOR_BGR2RGB)
        resized_img = cv2.resize(self.origin_img, (self.img_size, self.img_size))
        norm_img = resized_img / 255.0
        norm_img = (norm_img - self.mean) / self.std
        tensor_img = norm_img.transpose(2, 0, 1)
        tensor_img = torch.tensor(tensor_img, dtype=torch.float32).unsqueeze(0)
        vis_pre_img = np.clip((norm_img * self.std + self.mean) * 255, 0, 255).astype(np.uint8)
        return tensor_img, resized_img, vis_pre_img

class MCCA_DRA(nn.Module):
    def __init__(self):
        super().__init__()
        self.backbone = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=3, stride=2, padding=1), nn.ReLU(inplace=True),
            nn.Conv2d(32, 64, kernel_size=3, stride=2, padding=1), nn.ReLU(inplace=True),
            nn.Conv2d(64, 64, kernel_size=3, stride=2, padding=1), nn.ReLU(inplace=True)
        )
        self.pyramid_conv1 = nn.Conv2d(64, 32, kernel_size=1)
        self.pyramid_conv2 = nn.Conv2d(64, 32, kernel_size=3, padding=1)
        self.pyramid_conv5 = nn.Conv2d(64, 32, kernel_size=5, padding=2)
        self.scale_weights = nn.Parameter(torch.ones(3, dtype=torch.float32))
        self.rpn_score = nn.Conv2d(96, 1, kernel_size=3, padding=1)
        self.rpn_bbox = nn.Conv2d(96, 4, kernel_size=3, padding=1)

    def forward(self, x):
        base_feat = self.backbone(x)
        p1 = self.pyramid_conv1(base_feat)
        p2 = self.pyramid_conv2(base_feat)
        p3 = self.pyramid_conv5(base_feat)
```

```
ab) PS C:\Users\26819\AppData\Local\Programs\Microsoft VS Code> pip install customtkinter
loading customtkinter-5.2.2-py3-none-any.whl.metadata (677 bytes)
ting darkdetect (from customtkinter)
loading darkdetect-0.8.0-py3-none-any.whl.metadata (3.6 kB)
emont already satisfied: packaging in d:\anaconda\envs\deeplab\lib\site-packages (from customtkinter) (25.0)
ading customtkinter-5.2.2-py3-none-any.whl (296 kB)
ading darkdetect-0.8.0-py3-none-any.whl (9.0 kB)
ing collected packages: darkdetect, customtkinter
sfully installed customtkinter-5.2.2 darkdetect-0.8.0
ab) PS C:\Users\26819\AppData\Local\Programs\Microsoft VS Code & D:\anaconda\envs\deeplab\python.exe c:\Users\26819\Desktop\可
conda\envs\deeplab\lib\site-packages\customtkinter\windows\widgets\core_widget_classes\ctk_base_class.py:179: UserWarning: CTKLabel
use CTKImage instead.
```

番茄视觉精准分割系统 - 智能分析中枢

TomatoVision

多尺度图像精准分割引擎 v2.0

• 载入图像

• 开始分割

📄 保存结果

流水线状态

图像处理 (Pre)	待命
特征提取 (MCCA)	待命
边缘强化 (EE-FRC)	待命
联合推断 (SURL)	待命

> --- 系统初始化完成 ---
> 运行设备检测: CUDA
> 已检测
best_model.pth, 使用随机初始化权重
> 系统空闲, 等待输入图像...

1. 原始输入图像

Waiting for input...

2. 特征归一化视图

Waiting for input...

4. 边缘强化掩码

waiting for input...

5. 亚像素边缘增强

Waiting for input...

行 17, 列 33 | 空格: 4 | UTF-8 | CR/LF | Python | 已安装 | deeplab (3.12.12)